

Maximizing Productivity of CHO Cell-Based Fed-Batch Culture Using Chemically Defined Media Conditions and Typical Manufacturing Equipment

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A highly productive chemically defined fed-batch process was developed to maximize titer and volumetric productivity for Chinese hamster ovary cell-based recombinant protein manufacturing. Two cell lines producing a recombinant antibody (cell line A) and an Fc-fusion protein (cell line B) were used for development. Both processes achieved product titers of 10 g/L on day 18 under chemically defined conditions. For cell line B, the use of plant derived hydrolysates combined with the optimized chemically defined medium increased the titer to 13 g/L. Volumetric productivities were increased from a base line of about 200 mg/L/d to about 500 mg/L/d under chemically defined conditions and as high as 700 mg/L/d with cell line B using plant derived hydrolysates. Peak cell densities reached greater than 20×10^6 v/c/mL, and cell viabilities were maintained above 80% on day 18 without the use of antiapoptotic genes or temperature shift. A rapid compound screening method was developed to effectively test positive factors within 72 h. Peak volumetric oxygen uptake rates (OUR) more than tripled from the baseline condition. Oxygen demand continued to increase after maximum cell density was reached with a maximal OUR of 3.7 mmol/L/h. The new process format was scaled up and verified at 100 L pilot scale using reactor equipment of similar configuration as used at manufacturing scale. © 2010 American Institute of Chemical Engineers Biotechnol. Prog., 26: 1400–1410, 2010

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Introduction

Manufacturing of recombinant protein products using mammalian cell culture technology has made significant progress in the last 2 decades. This development was, in particular, driven by the revolution in antibody technologies and related dose and cost of goods requirements.^{1,2} Monoclonal antibodies and related molecules have become a major therapeutic modality for indications in areas of inflammatory disorders, oncology, neurology, and infection.^{3–6} Chinese hamster ovary (CHO) or murine myeloma cell lines (NS0, SP2/0) are typically used for recombinant protein production, especially monoclonal antibodies.^{7–8}

Both fed-batch and perfusion processes have been established at industrial scale.^{7,9} The benefits of fed-batch processing are fast development, simple format, ease of scale-up, and compatibility with large scale manufacturing of up to 20,000 L scale.^{10,11} Benefits of perfusion processing are high-volumetric output because of typically high-cell densities and short product exposure to cultivation conditions. The short product residence time makes perfusion technology particularly attractive for sensitive products, such as blood clotting factors or other proteins undergoing proteolytic attack or other forms of degradation under culture conditions.^{12–14} For relatively stable products such as mono-

clonal antibodies, fed-batch processing has become the dominant production technology at industrial scale.

One of the remarkable improvements over the last 2 decades for mammalian cell culture at industrial scale is the increase of product concentration (titer). Product concentrations were typically in the range of 1–100 mg/L in 1980s, and the first generation of industrial fed-batch manufacturing processes for monoclonal antibodies developed in the 1990s typically featured titers in the range of up to 1 g/L with run durations of about 10–14 days.⁹ The volumetric productivity of such processes was typically in the range of 50–100 mg/L/d. Such first generation processes usually benefited from the use of animal derived and later nonanimal-derived hydrolysates to fortify media and feeds for higher cell densities and improved productivity.¹⁵ Nowadays, 1–5 g/L product concentrations or about 200 mg/L/day in volumetric productivity are becoming the typical expectation.^{9,10}

Medium optimization is considered as one of the key contributors to the increase of titer. Over the last 10 years much progress was made in understanding cell nutrient requirements and the development of chemically defined media and feed formulations to better control cell metabolism and to avoid the use of animal-derived raw materials and variability associated with ill-defined hydrolysates. Screening of new small molecule compounds as medium additives and balancing of current medium components by studying their interactions have been reported to increase titer and productivity.^{16,17} Even though non-animal derived hydrolysates are

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still widely used, chemically-defined culture conditions have become more and more common.

In addition to medium optimization, proliferation control strategies have been investigated to reduce or inhibit apoptotic processes to achieve improved culture longevity, and thus, enhanced productivity. These include temperature shift to conditions of mild hypothermia,^{18,19} the use of apoptosis inhibitors²⁰ and cell engineering-based approach to inhibit apoptosis.^{21,22}

A different approach to increase product concentrations at the end of a culture is the use of an ultrafiltration membrane for concentrating the product while perfusing the culture. This principle was commonly used in the 1980s with hollow fiber bioreactors,^{23,24} which has yielded product concentrations of up to 16 g/L for an antibody produced using a transformed B-cell.²⁵ Although such an approach does increase the product concentration in the bioreactor, it does not necessarily increase overall facility output as it merely concentrates the product similar to an ultrafiltration step applied after harvest. In contrast, increased product concentrations in fed-batch or regular perfusion culture (using product permeable perfusion devices) will increase product concentration and facility output.

In this article, we demonstrate a specific approach taken to increase cell mass, enhance culture longevity, while simultaneously improving cell specific productivity. A rapid screening method was developed to effectively screen components during the late growth phase of fed-batch culture. The extensive list of screening compounds and their effects on CHO cells are beyond the scope of this article, and are reserved for future publication. The improved high-yield process was further characterized through oxygen uptake rate (OUR) measurements to aid the understanding of process scalability. Finally, we discuss aspects of implementing such high-density fed-batch mammalian cell culture processes in typical manufacture bioreactors.

Materials and Methods

Cell lines

The two cell lines used in this study were generated using a DG44 CHO host that had been adapted to growth in serum-free medium.²⁶ The expression system used is based on DHFR amplification.^{27,28} Cell line A produced a nonglycosylated human monoclonal antibody. Cell line B expressed an Fc fusion protein.

Culture media

Three basal media (CM1, CM2, and CM3) and four feed media (CF1, CF2, CF2a, CF2b) were used in this study. Basal medium CM1 is a proprietary medium derived from IMDM/MCDB (1:1) with enrichment in amino acids and trace elements.^{29,30} Basal media CM2 and CM3 are derived from CM1 by addition of hydrolysate (CM2) and 1 g/L Pluronic F68 (CM3), respectively. CM2 was used as a baseline production medium in clinical manufacturing, and CM3 was used for all the other bioreactor productions under chemically defined medium conditions. CF1 feed medium was derived from a partial concentrate of CM1 with the removal of inorganic salts. Amino acid concentrations were determined by calculation of uptake rates of previous platform processes supplemented with hydrolysates. CF2 feed medium contained an approximately two-fold higher amino acid concentrations compared with CF1 and additional trace metal

supplementation. CF2a and CF2b are further modifications of CF2 with the addition of the positive factors identified through rapid component screening method.

Cell culture methods

Cell lines were thawed and maintained in CM1 medium with insulin (Millipore, MA) as the sole growth factor and were passaged in 1-L shake flasks (Corning, NY) every 2–3 days. For scale down experiments that mimic cells in production mode, cells were inoculated into either 250-mL or 1-L shake flask on a 50-mm orbital diameter platform shaker-incubator (Infors AG, Switzerland) controlled at 36°C, 5% CO₂, 150 rpm, and 85% humidity to avoid evaporation.

Bioreactor culture conditions

Bench Scale bioreactor production studies were conducted using 5-L glass bioreactors equipped with Applikon ADI 1060 Biocontrollers (Applikon Biotechnology, CA) or Sartorius B-DCU controllers (Sartorius AG, Germany). The temperature was kept at 36°C. The pH set point was controlled at 7.0 for cell line A and 7.1 for cell line B through addition of CO₂ and 1.0 M Sodium Carbonate. Dissolved oxygen (DO) was controlled through the delivery of both air and oxygen. Agitation was kept at 200 to 350 rpm. Surface aeration was controlled at 0.01 vvm for the first 3 days and then increased to 0.04 vvm for the remainder of the run. Feed was administered every 24 or 48 h starting day 3; the overall feed volume was proportional to the integral of viable cells (IVC) of the corresponding process.

Offline analytical methods

Bioreactor sampling was performed daily to measure metabolic parameters including glucose, glutamine, glutamate, lactate, ammonium, sodium, and potassium using the NOVA Bioprofile 400 Analyzer (NOVA Biomedical, WA). Cell density and viability were measured by means of Trypan blue exclusion using an automatic Cedex Cell Counter (Innovatis AG, Germany). pH and pCO₂ were measured using NOVA pHOX Analyzer (NOVA Biomedical, WA). Osmolality was measured using Auto Osmometer (Advanced Instruments, MA). Lactate dehydrogenase (LDH) was measured during cell damage experiments using a previously described method.³¹ Supernatant samples were sterile filtered and stored at 2–8°C for further titer and product quality analyses. The cumulative cell growth or IVC is determined from the area under the viable cell density curve and is estimated by using a sum of trapezoids approximation across the desired time interval.

Analysis of protein concentration

The antibody concentration was measured using an HPLC system (Waters, MA) with a UV detector, and a protein G affinity column (Applied Biosystems, CA). A Polychem analyzer (Polymedco, NY) was used to measure IgG concentration for faster turnaround time in rapid screening experiments. The specific productivity q_p (pg/cell/day, or pcd) is determined by dividing the change in titer over IVC during the same time interval. Daily production rate (mg/L/day) is calculated by dividing the change in titer over the time (in days) between two sample intervals. The average volumetric productivity, Q_p , (mg/L/day) or the average

amount of product produced per day can be described by the following equation:

$$Q_p = P/D \quad (1)$$

where P (mg/L) is the product titer at a certain run time (D) in days.

Analysis of amino acids

Quantification of amino acids was performed using an HPLC system (Agilent, CA), using a modified OPA/ FMOc derivatization method.³² The derivatized amino acids are separated by reverse phase HPLC.

Analysis of product quality

The percentages of intact antibody, aggregation, acidic variant, and glycosylation were measured from time course cell culture samples. Percentage of intact antibody was analyzed using nonreducing gel chip (Agilent 2100 Bioanalyzer and Protein 200 plus Kit). Percentage aggregation was measured using size exclusion chromatography. Imaging capillary isoelectric focusing (ICIEF) was used to measure the acidic variant levels of the product. GelChip-Capillary electrophoresis (Agilent, CA) was used to measure the glycosylation of fusion protein for cell line B.

OUR modeling

OUR was estimated using the oxygen transfer rate (OTR) by applying the stationary liquid phase balance method.³³ At steady state, dC/dt is zero, and OUR can be assumed to be equal to OTR based on the mass transfer Eq. (2):

$$OUR = OTR = k_L a(C^* - C) \quad (2)$$

where $k_L a$ is the mass transfer coefficient, C^* is the oxygen saturation concentrations in the liquid phase, and C is the oxygen concentration in the liquid phase measured through a DO probe. $k_L a$ was measured through the dynamic method and correlated to power input per unit volume and superficial gas velocity. The correlation can further be simplified to the Eq. (3) for the ease of application

$$k_L a = C \times (\text{rpm})^\alpha (\text{mlm})^\beta \quad (3)$$

where rpm stands for impeller agitation speed (rotation per minute), and mlm represents total sparge rate at ml per minute. During fed-batch cultivation, the culture volume changes over time because of feed addition. Two correlations were established to better estimate $k_L a$ values (h^{-1}) in 5 L baffled bioreactors. At 3 L working volume, $k_L a$ is empirically measured and defined as

$$k_L a = 1.95 \times 10^{-6} \times (\text{rpm})^{2.10} (\text{mlm})^{0.50} \quad (4)$$

At 2.5L working volume, the correlation is

$$k_L a = 1.53 \times 10^{-7} \times (\text{rpm})^{2.68} \times (\text{mlm})^{0.53} \quad (5)$$

Rapid component screening method

A rapid component screening method was developed in shake flasks to provide an efficient screening platform. Figure 1a shows a comparison of the workflow in the con-

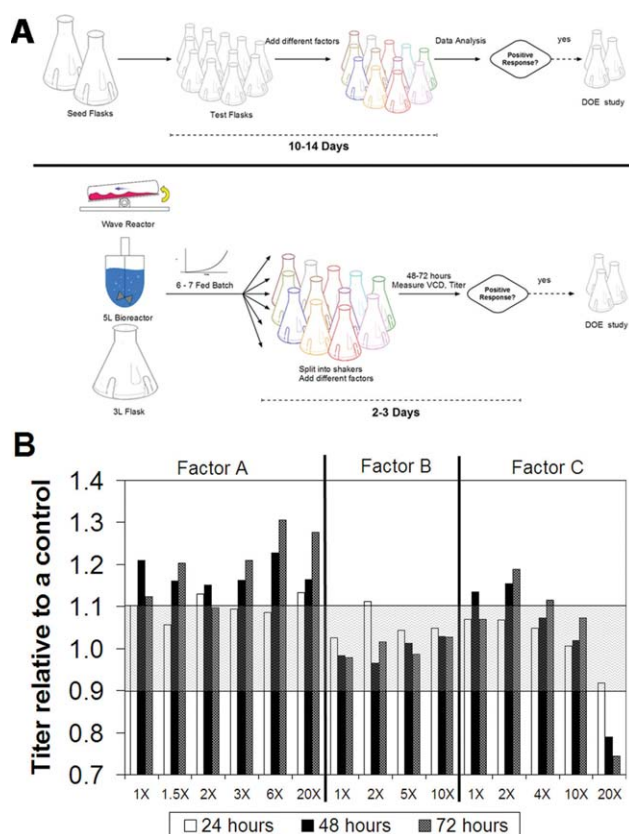


Figure 1. Rapid component screening method.

A: Comparison of a conventional component screening workflow (top) vs. the workflow used here (bottom). B: Increase in titer over control for three typical response categories. Shaded area indicates "insignificant" response to account for experimental errors. 1× indicates starting concentration, 20× indicates 20 folds of starting concentration.

ventional vs. rapid screening method. Cells were initially seeded in either a 3 L shake flask, 5 L bioreactor, or a 10 L wave bioreactor. Feed medium (CF1 or CF2) was added to promote cell growth to a density of over 10^6 viable cells/mL before splitting into multiple 250 mL shake flasks supplemented with various concentrations of tested components. Samples were withdrawn 48 and 72 h later to measure cell density, viability, and protein concentration. Figure 1b demonstrated three categories of responses typically observed. Positive response, demonstrated by component A, was defined as greater than 10% improvement in protein concentration at the time points of 48 and 72 h with no negative impact on cell viability. Component B, represents over 60% of tested molecules, which failed to generate more than 10% improvement. The 10% limit is set to account for instrumentation error or data variability from Polychem IgG assay. Component C demonstrated a narrow therapeutic range of positive response, and was further verified for its optimal concentration.

Results

Base process performance

A fed batch process using plant hydrolysate (H-1 process) was initially developed for cell line A and scaled successfully to 2,000 L manufacturing scale. A typical 5 L bench scale bioreactor performance of H-1 process is shown in

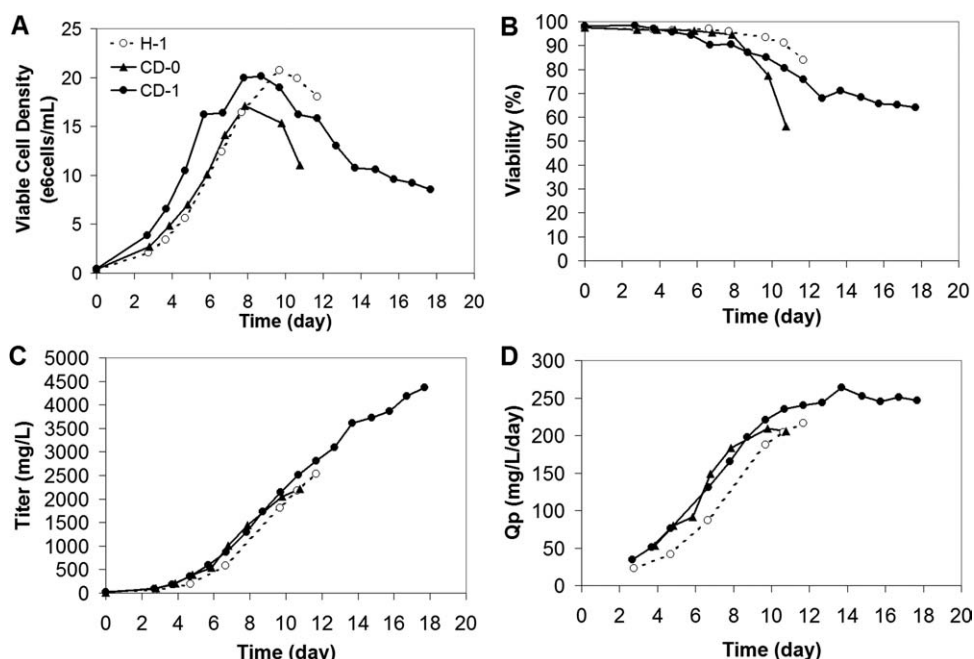


Figure 2. Culture performance of cell line A under three process conditions.

Time profiles of A: viable cell densities. B: viability profiles. C: titer D: Average volumetric productivity. H-1, hydrolysate containing base process. CD-0, H-1 process with omission of hydrolysate. CD-1, CD process after first round of media and feed optimization.

Figures 2A to D. The use of plant hydrolysate supports cell growth to about 20E6 cells/mL with a final titer of 2.5 g/L and harvest viability of 83% after 12 days. The process was used as a benchmark to develop an improved chemically defined process format.

Optimization of cell mass under chemically defined process conditions

CM1 (chemically defined) and CM2 (hydrolysate containing) culture media contained 1 g/L Pluronic F68, with similar surface tension around 53–55 dyn/cm. These values are in the range reported to significantly lower cell-to-bubble attachment.³⁴ However, when cell line A was cultured at 5 L scale in CM1 medium, a decline in cell viability was observed after aeration was initiated using a sintered sparger. The viability decline was significantly lower when using the hydrolysate containing CM2 medium. To investigate bubble-related cell damage under chemically defined conditions, three cultures were inoculated at 1E6 viable cells/mL and aerated with (a) 35 mL/min air through a 50 μ m sparger in CM3 medium (CM3 is CM1 with additional 1 g/L Pluronic F68), (b) condition a in CM1 medium, and (c) 15 mL/min air through a 15 μ m sparger in CM1 medium. Six hours later, cell viabilities dropped from 99 to 92, 86, and 81%, respectively. Supernatant LDH was measured and indicated lysis of about 0.18, 0.26, and 0.4 e6 cells/mL, respectively. This indicates that 2 g/L Pluronic F68 in CM3 alleviated cell damages from small bubbles generated from the 50 μ m sintered sparger. Therefore, 2 g/L Pluronic F68 in conjunction with a drilled hole sparger was chosen for development of the chemically defined process. These changes helped maintain high viability throughout the growth phase and increased process robustness.

The plant hydrolysate used in CM2 contained glutamine-rich polypeptides, and inevitably introduced higher ammonium into the process. By removing the hydrolysate from

production medium, the maximum ammonium concentration was lowered from 7.3 to 2.8 mM. However, elimination of the hydrolysate (CD-0 process) resulted in faster viability decline (Figure 2B) and shorter culture duration (11 days). Hydrolysates are known to boost cell mass and stabilize viability but specific effects of different hydrolysate components remain elusive. To restore cell growth and viability, amino acid balancing was performed first. Specific amino acid uptake rates under CD process conditions were measured and compared with the hydrolysate-containing process. The concentrations of both essential and nonessential amino acids in the feed were then adjusted in a way similar to the method described by Xie and Wang.³⁵ Second, a titration of trace elements in CF1 was conducted to identify their effect on cell viability. All trace elements were tested as a group without individually optimized. Figure 3A depicts an optimal range of trace elements for cell growth and productivity. Trace metal enrichment significantly enhanced cell viability but an overdose of trace metals can be toxic to cells (Figure 3B). Several water-soluble vitamins decreased to less than 25% of their initial concentrations at later stage of the fed-batch process. However, further supplementation of vitamins did not lead to significant increase in cell mass and cell viability, implying they may not be essential. Several rounds of iterations followed and resulted in stabilization of cell mass to about 18 days and 25% increase in peak cell densities compared to the baseline process. The overall cell mass after the first round of CD process optimization (CD-1 process) increased from 88 to 206E6 cells-day/mL. Volumetric productivity increased marginally to about 250 mg/L/d with a final titer of 4.5 g/L on day 18 (Figures 2C and D).

Optimization of specific productivity under chemically defined process conditions

After cell mass was stabilized and an overall 2.3 fold increase in IVC was achieved, attention was shifted to selective components in the existing medium, which may further

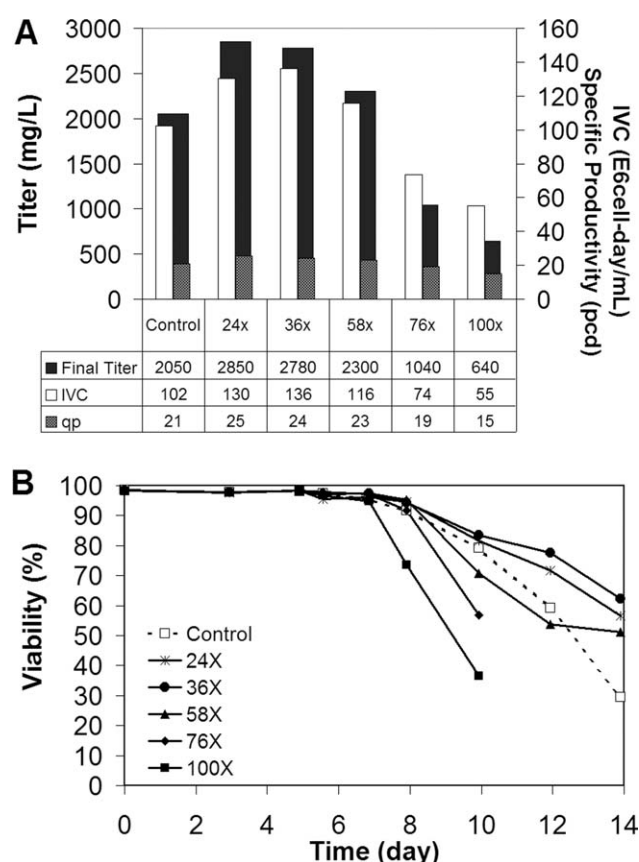


Figure 3. Effects of trace element supplementation in chemically-defined process.

A: Concentration dependent effects of elevated trace element concentrations on C-IVC, Titer, and specific productivity of cell line A. B: Viability profiles of trace element titration.

improve cell productivity. After the first round of CD optimization, a constant specific productivity of 21 pcd in both growth phase and production phase was observed (Figure 4F). The rapid component screening method was used to test a number of existing medium components in basal medium CM1.

Together with the increase in cell mass, ammonium levels were nearly doubled to 5.5 mM, which may negatively impact product quality and cell physiology.^{36–39} Positive factors identified from the screening method were incorporated into CF2a feed medium, which effectively inhibited ammonium accumulation in CD-2 process (Figure 4E) while maintaining the osmolality profile close to the baseline process.

In addition to optimizing existing medium components, new chemically-defined components were also explored for productivity enhancers, including various fatty acids, steroids, dipeptides, tripeptides, nucleotides, nucleosides, lipids, and so on. The efforts eventually led to the use of CF2b feed in the third round of CD optimization (CD-3 process). While all versions of CF2 feed media did not improve intrinsic cell productivity during the growth phase, they led to significant specific productivity increase in the stationary phase, from about 20 to over 40 pcd (Figure 4F). Table 1 summarizes process characteristics and their evolution from the baseline process to the improved processes. Overall, cell line A achieved a 240% increase in IVC and a 60% increase in specific productivity over the first generation (CD-0) process. The final titer reached an average of 9.8 g/L with a volumetric productivity of about 550 mg/L/day (Figures 5C and D).

Cell viability remained above 80% on day 18 (Figure 4B). The highest titer achieved for cell line A was 12 g/L on day 20 (data not shown).

Pilot scale verification

Confirmation of the bench scale results was performed at 100 L pilot scale. A total of three pilot scale runs were performed for cell line A to verify the scalability of the improved process format. Figure 5 shows the average growth, viability, titer, and volumetric productivity profile of three 100 L culture in comparison with the average of five batches done at 5 L scale. All performance parameters were comparable between the 5 L and 100 L scales.

Cell line B

A second cell line was used to verify performance enhancements gained with cell line A. Under CD-2 process, a titer of 11 g/L was achieved (Figure 6C). By adding a plant hydrolysate to the optimized CD process (H-2), the performance was enhanced to a titer of about 13 g/L. Volumetric productivities peaked at about 600 and 800 mg/L/d, respectively, (Figure 6D). Under both conditions, the maximum viable cell density was around 24E6 cells/mL, with final harvest viabilities above 70% on day 18 (Figures 6A and 6B).

OUR assessment

The OUR were obtained using the OTR model as described in Materials and Methods using cell line A. During the growth phase, OUR trended up as cell densities increased (Figure 7A). When the cultures progressed near peak cell density around day 8, the OUR of the CD-3 process continued to increase, and the specific OUR (qO_2) in the production phase was higher than that during the growth phase (Figure 7B). On the other hand, OUR of the baseline process with and without hydrolysates leveled off after peak cell density was achieved at ~33 and 60% of maximum OUR seen with the optimized CD-3 process (Figure 7A). The increase in volumetric OUR correlated well with the increase in daily production rate of the CD-3 process (Figure 7C).

Product quality assessment

Figure 8A summarizes several product quality attributes of the antibody produced by cell line A. In terms of % intact antibody and % aggregation, no significant changes were seen over the course of production. Both % intact antibody and % aggregate were similar to the baseline process. However, an increase in acidic variants up to a maximum of 50% was observed. The increasing low pI isoform may be attributed to deamidation, which is expected under extended culture conditions. Figures 8B and C show distribution of G0, G1, and G2 glycans in the Fc portion of product B. A fairly typical distribution for Fc fusion proteins was detected with mainly G1 and G0 structures. Between days 10 and 18 the galactose content decreased slightly over time. Higher galactosylation is generated through CD-2 process.

Discussion

In this study, we used a rapid component screening method conducted in shake flasks at high-cell density. We

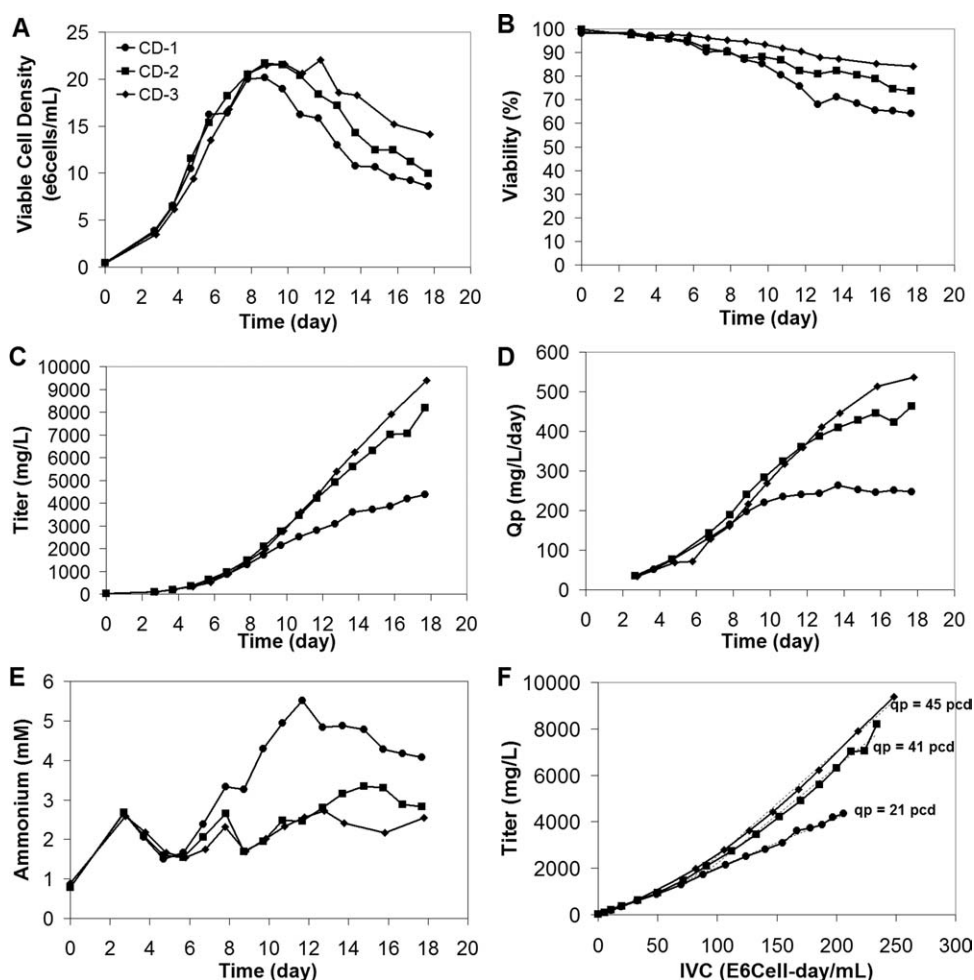


Figure 4. Improved culture performance for cell line A during three rounds of CD optimization.

A: Viable cell density. B: Viability. C: Titer D: Average volumetric productivity. E: Ammonium concentration. F: Titer vs. IVC profile. Specific productivity (q_p) indicated was calculated from the slopes of each culture in the stationary phase (IVC range of 60 to 250E6 cell-day/mL).

chosed shake flasks due to the practicality as well as good and predictable growth characteristics for our cell lines. It is crucial to avoid oxygen limitation in a screening platform, as suboptimal culture conditions might generate misleading results.⁴⁰ Optimization of culture volume, agitation rate, and the pH buffer system resulted in a pH profile that ensures acceptable growth without the need for base addition. This enabled the use of cultures from late exponential growth at high-cell density that are representative of those cultivated in the bioreactors. In addition, cultures at late exponential phase consistently demonstrated greater sensitivity to positive factors at 48 and 72 h than cultures in early growth phase or stationary phase.

New components identified through the screening method and applied to the CF2b feed formula were shown to improve specific productivity for cell line A. Not all of the components identified show the same effects on cell line B. It is not uncommon that many productivity enhancers are cell line specific.¹⁶ We hypothesize that this result may be related to underlying limitations for expression of the respective protein products. Whereas cell line A had an intrinsic specific productivity of about 20 pcd during the growth phase, and cell line B was more productive with about 44 pcd under the same condition. The specific productivity for cell line A was improved to about 45 pcd in the CD-3 process after peak cell density was reached, while specific

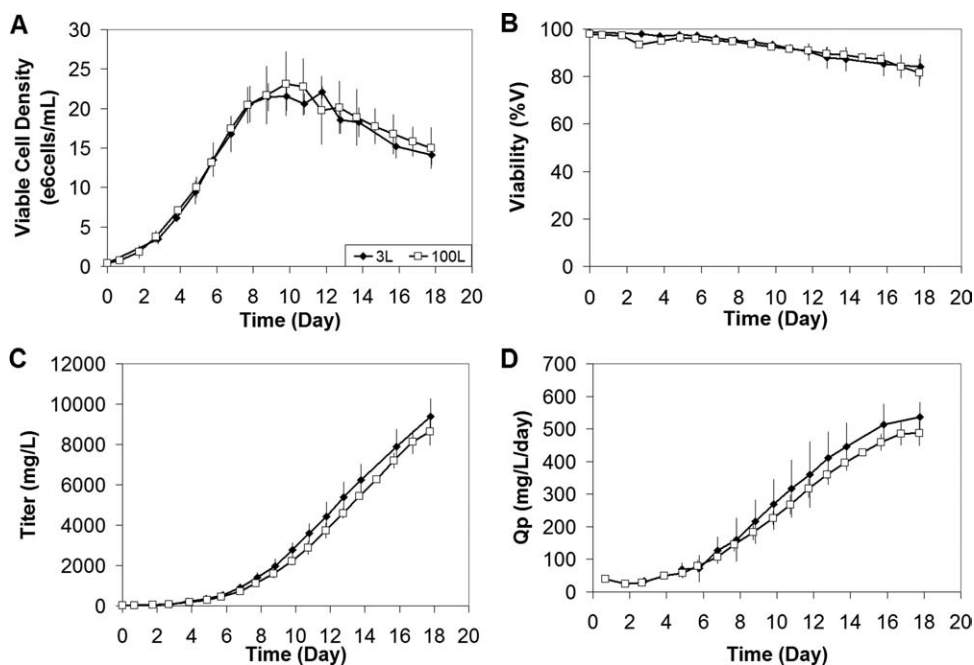
productivity remain the same for cell line B during growth and stationary phases. The introduction of a plant hydrolysate in the H-2 process was able to further improve the q_p for cell line B to about 54 pcd. Under this culture condition galactosylation was lower compared with the CD-2 process (36 vs. 27%).

The main change in product quality observed as a function of the new process format and extended run duration was an increase in low pI isoforms (Figure 8). This is not unusual and typically seen when protein is incubated under cultivation conditions. The main driver for low pI isoforms on antibodies is deamidation of asparagine residues. Because of the nature of this nonenzymatic modification pH and temperature are known to be the main drivers.^{41,42} Lowering cultivation pH or temperature can help reduce deamidation if this becomes a problem,⁴³ but its impacts on the specific productivity and metabolite waste accumulation should be further investigated. In addition engineering primary amino acid sequence near deamidation sites can reduce sensitivity of particular asparagines.⁴⁴ However, such an approach would require protein engineering on a more elaborate time line.

To date, typical volumetric productivities for antibody fed-batch processes are in the range of 200 mg/L/day (2–3 g/L titers with run durations typically in the 10–15 day range). Here, we demonstrate volumetric productivities of over 500 mg/L/day, which is one of the highest yielding fed-

Table 1. Summary of Process Performance Parameters for Cell Line A and Cell Line B During Process Optimization Stages

	Cell Line A								Cell Line B	
	Hydrolysate	No hydrolysate	1st round	2nd round	3rd round	Pilot scale 01	Pilot scale 02	Pilot Scale 03	No hydrolysate	Hydrolysate
Process	H-1	CD-0	CD-1	CD-2	CD-3	CD-3	CD-3	CD-3	CD-2	H-2
Vessel size (L)	5	5	5	5	5	100	100	100	5	5
Basal Medium	CM1	CM2	CM3	CM3	CM3	CM3	CM3	CM3	CM3	CM3
Feed Medium	CF1	CF1	CF2	CF2a	CF2b	CF2b	CF2b	CF2b	CF2a	CF2c
Total run time (day)	12	11	18	18	18	18	18	18	18	18
Final Integral of viable cells (10^6 cells $\times$ day/mL)	150	90	210	240	250	220	280	270	250	270
Final cumulative specific productivity (pcd)	16	27	21	36	39	37	30	35	43	49
Final antibody yield (mg/L)	2,300	2,300	4,400	8,700	9,800	8,200	8,300	9,400	10,800	13,000
Harvest Viability (%)	83	56	64	74	84	83	75	87	80	72
Volumetric productivity (mg/L/day)	190	220	250	460	550	460	470	530	600	730
Peak viable cell density (10^6 cells/mL)	23	16	20	22	22	19	28	25	24	26
Peak lactate concentration (g/L)	2.3	2.4	2.5	2.4	2.4	2.3	2.4	2.7	2.5	3.8
Peak ammonium concentration (mmol/L)	7.3	2.8	5.5	3.4	2.6	4.3	3.1	3.2	8.7	7.4
Peak pCO ₂ (mmHg)	50	86	88	78	91	95	89	99	90	111
Peak osmolality (mOsm/kg)	360	340	410	390	380	400	370	350	410	540

**Figure 5. Comparison of cell culture performance between 5 L ($n = 7$) and 100 L ($n = 3$) cultures using cell line A.**

Time profile of A: viable cell density B: viability C: titer D: Average productivity profiles. Error bars indicate one standard deviation.

batch processes ever reported for mammalian cell culture using chemically defined medium. Such range of volumetric productivity is comparable with optimized perfusion processes. Konstantinov et al. (2006) proposed a “push-to-low” approach to improve the performance of high-density perfusion culture of animal cells for stable proteins.¹² They conclude that using a 20 pcd cell line at a low-cell specific perfusion rates (CSPR) of 0.05 nL/cell/day and high-cell density up to 80 E6/mL, volumetric productivities up to 1600 mg/L/day may be possible for optimized perfusion culture. In practice, at low perfusion rates there may be some difficulties in supporting cell growth to such high-cell density while maintaining high viability. Kompala and Ozturk (2007) showed that cell viability steadily declined from 80% to mid 50% when an enriched medium was used to increase

cell density to around 40E6/mL.⁴⁵ This problem may be solved by apoptosis engineering as reported by Nivitchanyong et al., 2007.⁴⁶ High viabilities maintained above 90% with a CSPR of 0.1 nL/cell/day were reported. Very high-cell densities in perfusion culture pose great challenges for cell retention devices during scale up, whereas fed-batch cultivation is conducted at 15,000 L scale routinely.¹¹ On the other hand, perfusion culture enables short product residence times, and thus, may be advantageous for unstable proteins that may not maintain adequate product quality in a fed-batch process. It has also been suggested that by using perfusion technology followed by a fed-batch production culture, the impact of this combination can boost cell mass and reduce metabolic waste in the production state.⁴⁷ Here we demonstrate that by careful media optimization, high cell

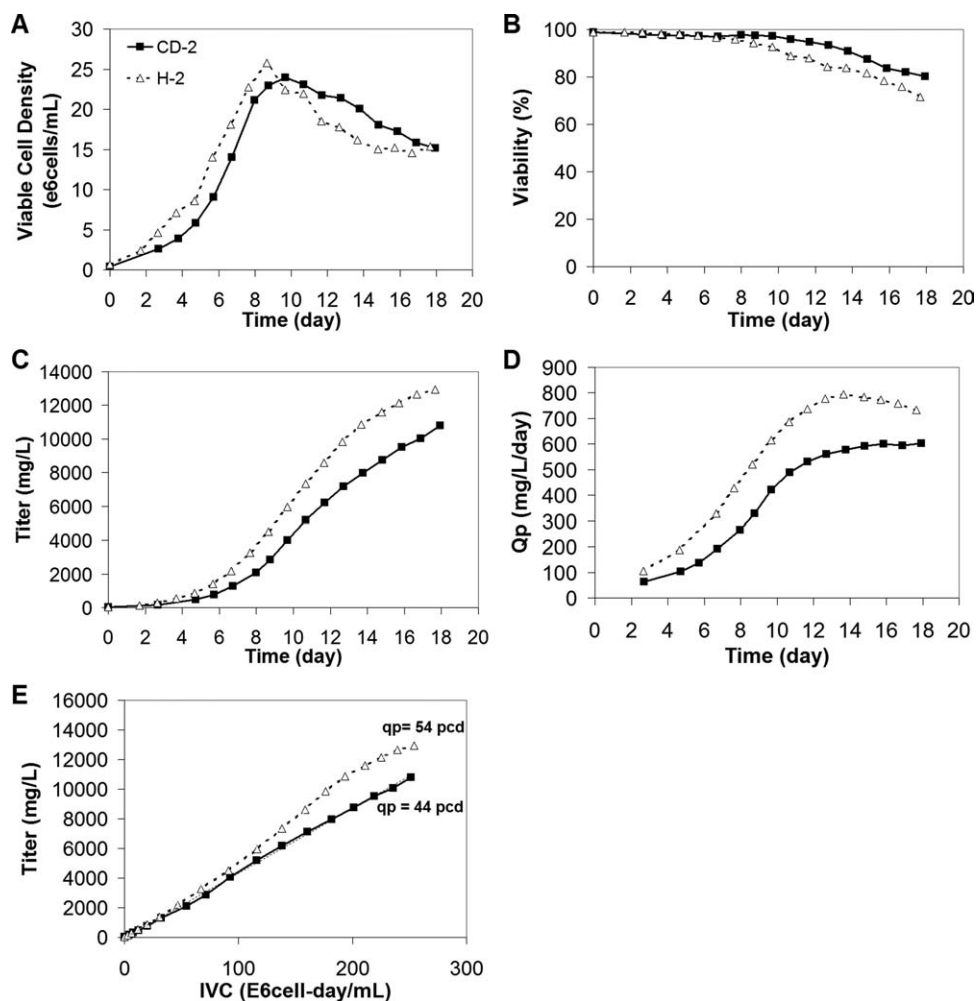


Figure 6. Comparison of average cell culture performance of cell line B under CD-2 ($n = 2$) and H-1 process conditions.

Time profiles of A: viable cell densities, B: viability C: titer D: Average volumetric productivity, E: Titer vs. IVC profile. Specific productivity (q_p) indicated was calculated from the slopes of each culture over the whole time course.

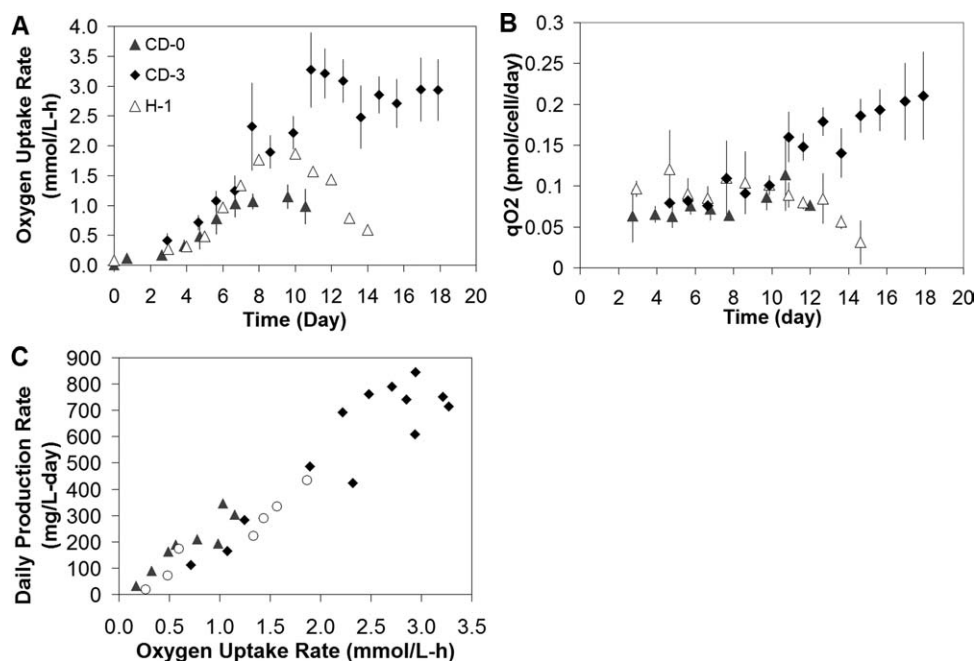


Figure 7. Comparison of OUR, q_{O_2} and daily production rates in three process conditions.

No hydrolysate (CD-0, $n = 3$, $\blacktriangle$), hydrolysate-containing (H-1, $n = 3$, $\circ$) and optimized CD process (CD-3, $n = 3$, $\blacklozenge$). Error bars represent one standard deviation of each data point. A: OUR vs. time profiles. B: Specific oxygen uptake rate vs. time profiles. C: OUR vs. daily production rate profiles.

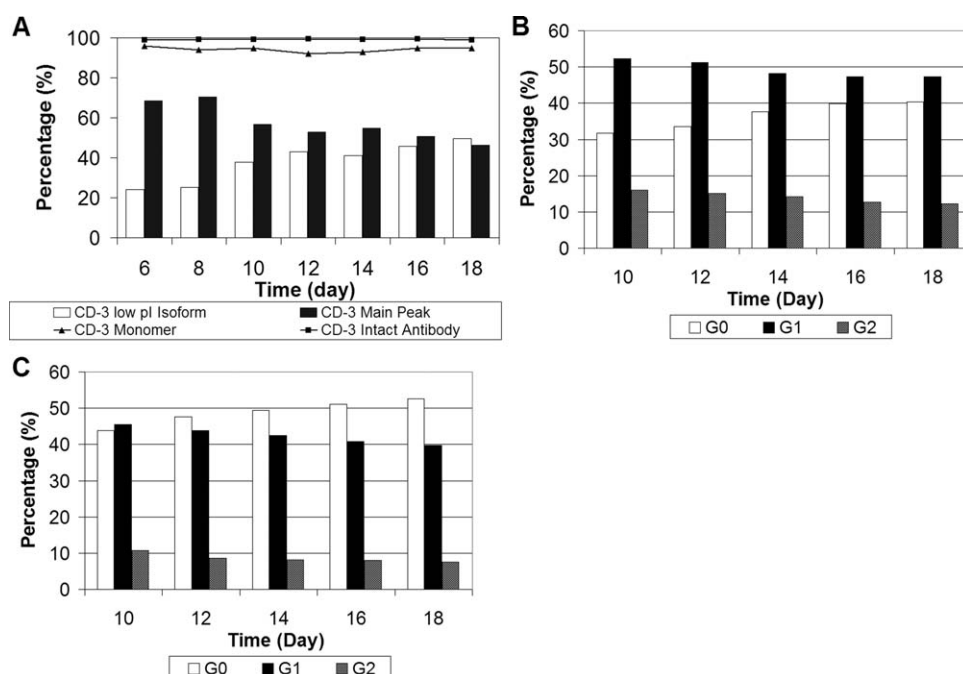


Figure 8. Product quality assessment.

A: Time profile (day 6 to 18) of product quality data for cell line A after the third round of CD optimization (CD-3 process). % low pI isoform and % main peak were determined by ion exchange chromatography. % Monomer was determined by size exclusion chromatography and % intact antibody was measured using nonreducing gel chip. B: Time course data of glycan distribution for cell line B in CD-2 process. C: Time course data of glycan distribution for cell line B in H-2 process.

mass and high productivity can be achieved without the use of perfusion system. Volumetric productivities of optimized fed-batch cultures are competitive to perfusion cultures for stable molecules, such as antibodies. The feed was administered without continuous addition or sophisticated feedback control at a feed volume ratio much lower than perfusion process required. Cell viability can be maintained at above 80% without the use of antiapoptosis genes or temperature shift in fed-batch process out to about 20 days. The slightly reduced cultivation temperature (36 vs. 37°C) may have slowed down cellular metabolism, and contributed to stable, yet high, cell productivity. Metabolic waste production was minimized, and osmolality was maintained at a level amenable for extended stationary phase.

High density mammalian cell culture presents some engineering challenges, especially in large-scale manufacture processes, where mass transfer may be a limiting factor. Typical stirred tank reactor systems used in the industry for mammalian cell culture featured mass transfer coefficients (k_{La}) in the order of 5–15 h⁻¹ depending on power per volume used and the aeration strategy applied.⁴⁸ When a culture is maintained at 30% air saturation with the use of pure oxygen, the OTR achievable is ~15 mmol/L/h, assuming maximum k_{La} of 15 h⁻¹. With a typical specific uptake rate for mammalian cells of about 0.3 pmol/cell/h, this suggests a maximal achievable cell concentration in the range of 50E6 cells/mL.

CO₂ removal becomes a challenge when a high percentage of oxygen is used for aeration. Because of relatively high solubility of CO₂ and a low Henry's law constant (pressure over liquid concentration), a significant portion of air in the inlet gas mixture may be required to remove enough CO₂ thus reducing the driving force for oxygen. Also, as the bubble residence times increase with scale, the average driving force for CO₂ removal further declines.⁴⁹ Elevated culture

pCO₂ can negatively impact growth, metabolism, and productivity in addition to adverse effects on glycosylation.^{50,51} High pCO₂ may also result in increased osmolality because of base addition for pH control. Therefore, the air cap in normal cascade mode for DO control needs to be carefully adjusted to balance the air to oxygen ratio based on the culture carbon dioxide evolution rate and maximal acceptable pCO₂ level. Mostafa and Gu (2003) reported a 1:1 ratio of air to oxygen aeration rate in a 1,000 L bioreactor run.⁵² Based on this ratio, the maximum OTR is around 8.7 mmol/L/h at a k_{La} of 15 h⁻¹. This is two times higher than the maximum OUR described in this work (3.7 mmol/L/h) and suggests that typical reactor equipment used in the industry is capable of satisfying oxygen demand for high-cell density processes such as the optimized CD process presented here.

Goudar et al. (2007) documented that NaHCO₃ from medium and base addition contributed the most to the culture pCO₂ level for a perfusion process.⁵³ For a fed-batch process, most CO₂ stem from cell respiration.⁵⁰ In the presented high-yielding CD process, a well-controlled lactate level as well as culture dilution by bicarbonate-free feed medium kept peak pCO₂ at an acceptable level of less than 120 mmHg. A temperature shift could be applied to support even higher cell densities using typical manufacturing equipment as low temperature can significantly reduce specific OUR of cells.^{18,54}

One of the interesting observations with the optimized CD process is that OUR continued to increase after peak VCD was reached and an increase in qO₂ in the stationary phase. An increase of average cell diameter from 15.7 to 18.5 μm was also observed; the increase in specific productivity during fed-batch cultivation was at least to some degree mediated by a larger cell size. OUR can be an important parameter to monitor cell metabolism. qO₂ was reported to be independent from cell density manipulated by dilution⁵⁴

and maintained at relatively constant level in most of exponential growth.⁵⁵ A decrease in OUR was considered as a clear sign of the end of exponential growth⁵⁶ and has been used as a tool to guide feed optimization.⁵⁷ As a result, many fed-batch processes calculated or measured maximum OUR at peak viable cell densities to facilitate scale up planning during tech transfer.⁵⁸ However, we demonstrated that it is necessary to monitor OUR into the stationary culture phase until the daily production rate declined to prevent underestimating maximum OUR requirement for bioreactor process.

Further optimization of volumetric productivity of fed-batch or perfusion culture through cell mass intensification will ultimately lead to more challenging manufacturing processes. Daily feeding may become necessary and significantly more manipulations may reduce success rates. This is another challenge for further intensification of mammalian production processes and in this context it is worth noting that improving cellular productivity can achieve the same result with less investment of resources and simpler manufacturing processes. Ultimately guidance of where to invest resources (cell line or process) to improve volumetric productivity depends on the baseline process and cell line available. A low productive cell line (e.g., <20 pcd for monoclonal antibody) offers significant opportunity to impact manufacturing efficiency by further improvements of cell specific productivity. Similarly, a fed-batch process delivering less than 100E6 cell-day/mL IVC offers significant opportunity to improve efficiency by cell mass improvements.

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Literature Cited

- Farid SS. Process economics of industrial monoclonal antibody manufacture. *J Chromatogr B Analyt Technol Biomed Life Sci.* 2007;848:8–18.
- Chadd HE, Chamow SM. Therapeutic antibody expression technology. *Curr Opin Biotechnol.* 2001;12:188–194.
- Stiegler G, Katinger H. Therapeutic potential of neutralizing antibodies in the treatment of HIV-1 infection. *J Antimicrob Chemother.* 2003;51:757–759.
- Souriau C, Hudson PJ. Recombinant antibodies for cancer diagnosis and therapy. *Expert Opin. Biol. Ther.* 2001;1:845–855.
- Hutas G. Ocrelizumab, a humanized monoclonal antibody against CD20 for inflammatory disorders and B-cell malignancies. *Curr Opin Invest Drug.* 2008;9:1206–1215.
- Dalakas MC. B cells as the therapeutic targets in autoimmune neurological disorders. *Nat Clin Pract Neurol.* 2008;4:557–567.
- Chu L, Robinson DK. Industrial choices for protein production by large-scale cell culture. *Curr Opin Biotechnol.* 2001;12:180–187.
- Barnes LM, Bentley CM, Dickson AJ. 2003. Stability of protein production from recombinant mammalian cells. *Biotechnol Bioeng.* 2003;81:631–639.
- Wurm FM. Production of recombinant protein therapeutics in cultivated mammalian cells. *Nat Biotechnol.* 2004;22:1393–1398.
- Birch JR, Racher AJ. Antibody production. *Adv Drug Deliv Rev* 2006;58:671–685.
- Meuwly F, Weber U, Ziegler T, Gervais A, Mastrangeli R, Crisci C, Rossi M, Bernard A, von Stockar U, Kadouri A. Conversion of a CHO cell culture process from perfusion to fed-batch technology without altering product quality. *J Biotechnol.* 2006;123:106–116.
- Konstantinov K, Goudar C, Ng M, Meneses R, Thrift J, Chuppa S, Matanguihan C, Michaels J, Naveh D. The “push-to-low” approach for optimization of high-density perfusion cultures of animal cells. *Adv Biochem Eng Biotechnol.* 2006;101:75–98.
- Vogel JH, Anspach B, Kroner KH, Piret JM, Haynes CA. Controlled shear affinity filtration (CSAF): a new technology for integration of cell separation and protein isolation from mammalian cell cultures. *Biotechnol Bioeng.* 2002;78:806–814.
- Ryll T, Dutina G, Reyes A, Gunson J, Krummen L, Etcheverry T. Performance of small-scale CHO perfusion cultures using an acoustic cell filtration device for cell retention: characterization of separation efficiency and impact of perfusion on product quality. *Biotechnol Bioeng.* 2000;69:440–449.
- Jan DC, Jones SJ, Emery AN, Al-Rubeai M. Peptone, a low-cost growth-promoting nutrient for intensive animal cell culture. *Cytotechnology.* 1994;16:17–26.
- Allen MJ, Boyce JP, Trentalange MT, Treiber DL, Rasmussen B, Tillotson B, Davis R, Reddy P. Identification of novel small molecule enhancers of protein production by cultured mammalian cells. *Biotech Bioeng.* 2008;100:1193–1204.
- Zhang J, Robinson D, Salmon P. A novel function for selenium in biological system: selenite as a highly effective iron carrier for Chinese Hamster Ovary cell growth and monoclonal antibody production. *Biotech Bioeng.* 2006;96:1188–1197.
- Moore A, Mercer J, Dutina G, Donahue CJ, Bauer KD, Mather JP, Etcheverry T, Ryll T. Effect of temperature shift on cell cycle, apoptosis and nucleotide pools in CHO cell batch cultures. *Cytotechnology.* 1997;23:47–54.
- Kaufmann H, Mazur X, Fussenegger M, Bailey JE. Influence of low temperature on productivity, proteome and protein phosphorylation of CHO cells. *Biotech Bioeng.* 1999;63:573–582.
- van de Goor J. Improvement of industrial cell culture processes by caspase-9 dominant negative and other apoptotic inhibitors. *Cell Eng.* 2004;4:211–221.
- Chiang GG, Sisk WP. Bcl-x(L) mediates increased production of humanized monoclonal antibodies in Chinese hamster ovary cells. *Biotechnol Bioeng.* 2005;91:779–792.
- Al-Rubeai M, Singh RP. Apoptosis in cell culture. *Curr Opin Biotechnol.* 1998;16:67–85.
- Altshuler GL, Dziewulski DM, Soweck JA, Belfort G. Continuous hybridoma growth and monoclonal antibody production in hollow fiber reactors-separators. *Biotechnol Bioeng.* 1986;28:646–58.
- Knazek RA, Wu YW, Aebersold PM, Rosenberg SA. Culture of human tumor infiltrating lymphocytes in hollow fiber bioreactors. *J Immunol Methods.* 1990;127:29–37.
- Lehmann J, Bunttemeyer H, Jager V. Bulk culture of animal cells for antibody production: a comparison of reactors. *Food Biotechnol.* 1990;4:423–431.
- Prentice HL, Ehrenfels BN, Sisk WP. Improving performance of mammalian cells in fed-batch processes through “bioreactor evolution.” *Biotechnol Prog.* 2007;23:458–464.
- Pallavicini MG, DeTeresa PS, Rosette C, Gray JW, Wurm FM. Effects of methotrexate on transfected DNA stability in mammalian cells. *Mol Cell Biol.* 1990;10:401–404.
- Gandor C, Leist C, Fiechter A, Asselbergs FAM. Amplification and expression of recombinant genes in serum-independent Chinese hamster ovary cells. *FEBS Lett.* 1995;377:290–294.
- Iscove NN, Melchers F. Complete replacement of serum by albumin, transferrin, and soybean lipid in cultures of lipopolysaccharide-reactive B lymphocytes. *J Exp Med.* 1978;147:923–933.
- Ham RG, McKeenhan, WL, Jakoby WB, Pastan IH. Methods in enzymology cell culture. In: *Media and growth requirements*. San Diego: Academic Press. 1979; 44–93.
- Ma N, Koelling KW, Chalmers JJ. Fabrication and use of a transient contractional flow device to quantify the sensitivity of mammalian and insect cells to hydrodynamic forces. *Biotechnol Bioeng.* 2002;80:428–437.
- Henderson JW, Ricker RD, Bidlingmeyer BA, Woodward C. *Rapid, Accurate, Sensitive and Reproducible Analysis of Amino Acids*. Agilent Publication Number 5980–1193EN. Agilent Technologies, Palo Alto, CA; 2000.

33. Ruffieux P, Stockar UV, Marison IW. Measurement of volumetric (OUR) and determinative of specific (qO_2) oxygen uptake rates in animal cell cultures. *J Biotechnol*. 1998;63:85–95.
34. Hu W, Rathman JJ, Chalmers JJ. An investigation of small-molecule surfactants to potentially replace pluronic F-68 for reducing bubble-associated cell damage. *Biotechnol Bioeng*. 2008;101:119–127.
35. Xie L, Wang DI. Stoichiometric analysis of animal cell growth and its application in medium design. *Biotechnol Bioeng*. 1994;43:1164–1174.
36. Ryll T, Valley U, Wagner R. Biochemistry of growth inhibition by ammonium ions in mammalian cells. *Biotechnol Bioeng*. 1994;44:184–193.
37. Gawlitzek M, Papac DI, Sliwowski MB, Ryll T. Incorporation of 15N from ammonium into the N-linked oligosaccharides of an immunoadhesin glycoprotein expressed in Chinese hamster ovary cells. *Glycobiology*. 1999;9:125–131.
38. Gawlitzek M, Ryll T, Lofgren J, Sliwowski MB. Ammonium alters N-glycan structures of recombinant TNFR-IgG: degradative versus biosynthetic mechanisms. *Biotechnol Bioeng*. 2000;68:637–646.
39. Yang M, Butler M. Effect of ammonia on the glycosylation of human recombinant erythropoietin in culture. *Biotech Prog*. 2000;16:751–759.
40. Zimmermann HF, Anderlei T, Buchs J, Binder M. Oxygen limitation is a pitfall during screening for industrial strains. *Appl Microbiol Biotechnol*. 2006;76:1157–1160.
41. Cleland, JL, Powell MF, Shire SJ. The development of stable protein formulations: a close look at protein aggregation, deamidation, and oxidation. *Critic Rev Therapeut Drug Carrier Sys*. 1993;10:307–377.
42. Stratton, LP, Kelly RM, Rowe J, Shively JE, Smith DD, Carpenter JF, Manning MC. Controlling deamidation rates in a model peptide: effect of temperature, peptide concentration, and additives. *J Pharm Sci*. 2001;90:2141–2148.
43. Bui T, Hegde V, Rajendran S, Mandavia D, Albarghouti M, Chen CC. Control of deamidation of monoclonal antibodies in mammalian cell culture process—a case study. 238th ACS National Meeting. August, Washington, DC, USA; 2009.
44. Robinson NE, Robinson AB. Deamidation of human proteins. *Proc Natl Acad Sci USA*. 2001;98:12409–12413.
45. Kampala DS, Ozturk SS. Optimization of high cell density perfusion bioreactor. *Biotechnol Bioprocess*. 2007;30:287–416.
46. Nivitchanyong T, Martinez A, Ishaque A, Murphy JE, Konstantinov K, Betenbaugh MJ, Thrift J. Anti-apoptosis genes Aven and E1B-19K enhance performance of BHK cells engineered to express recombinant factor VIII in batch and low perfusion cell culture. *Biotech Bioeng*. 2007;98:825–841.
47. Hiller G. *Use of short duration perfusion to enhance productivity in fed-batch bioreactors*. Cell Culture Engineering XI, Queensland, Australia; 2008.
48. Nienow AW. Reactor engineering in large scale animal cell culture. *Cytotechnology*. 2006;50:9–33.
49. Meier SJ. Modeling carbon dioxide mass transfer in cell culture reactors: effects of bubble size and reactor height and implications for scale up. 221st ACS National Meeting. San Diego, CA, USA; 2001.
50. Zhu MM, Goyal A, Rank DL, Gupta SK, VandenBoom T., Lee SS. Effects of elevated pCO₂ and osmolality on growth of CHO cells and production of antibody-fusion protein B1: a case study. *Biotechnol Progress*. 2005;21:70–77.
51. Takuma S, Hirashima C, Piret JM. Dependence on glucose limitation of the pCO₂ influences on CHO cell growth, metabolism and IgG production. *Biotechnol Bioeng*. 2007;97:1479–1488.
52. Mostafa SS, Gu X. Strategies for improved dCO₂ removal in large-scale fed-batch cultures. *Biotechnol Progress*. 2003;19:45–51.
53. Goudar CT, Matanguihan R, Long E, Cruz C, Zhang C, Piret JM, Konstantinov K. Decreased pCO₂ accumulation by eliminating bicarbonate addition to high cell-density cultures. *Biotechnol Bioeng*. 2007;96:1107–1117.
54. Jorjani P, Ozturk SS. Effects of cell density and temperature on oxygen consumption rate for different mammalian cell lines. *Biotechnol Bioeng*. 1999;64:349–356.
55. Ramirez OT, Mutharasan R. Cell cycle and growth phase dependent variations in size distribution, antibody productivity, and oxygen demand in hybridoma cultures. *Biotechnol Bioeng*. 1990;36:839–848.
56. Schoenherr I, Stapp T, Ryll T. A comparison of different methods to determine the end of exponential growth in CHO cells cultures for optimization of scale up. *Biotechnol Prog*. 2000;16:815–821.
57. Zhou W, Chen C, Buckland B, Aunins J. Fed-batch culture of recombinant NS0 myeloma cells with high monoclonal antibody production. *Biotechnol Bioeng*. 1997;55:783–792.
58. Xing Z, Kenty BM, Li ZJ, Lee SS. Scale-up analysis for a CHO cell culture process in large-scale bioreactors. *Biotechnol Bioeng*. 2009;103:733–746.

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